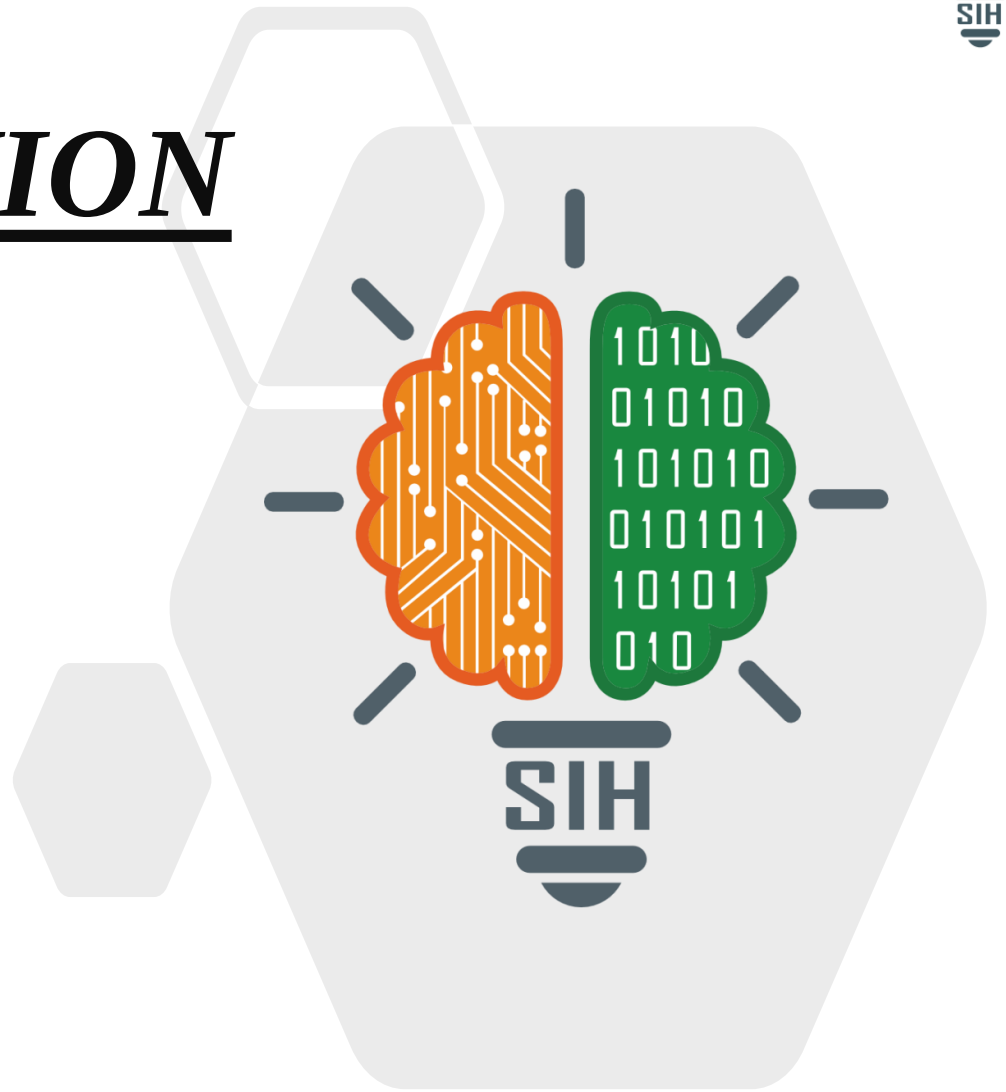
SIH INTERNAL HACKFEST 2026

SMART INDIA HACKATHON 2026

ROVION



- **Theme-** Disaster Management / Robotics and Drones
- **Category- Hardware**
- **Team Name - NEXORA**

NEXORA

Semi-Autonomous Disaster Rescuer

SMART INDIA HACKATHON 2026

❖ Proposed Solution :

- Drone scans hazardous zones & detects survivors using AI.
- Location is shared with the ground rescue rover.
- Rover approaches & confirms the survivor using sensors

❖ How It Addresses the Problem:

- Faster search across large disaster zones.
- Safer rescue by reducing human exposure.
- Real-time location & data for rescue teams.

❖ Innovation and uniqueness of the solution :

Drone detection → Target localization → Target handoff → Rover approach → Close-range confirmation

NEXORA

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

TECHOCLOGIES TO BE USED

DRONE


Pixhawk


GPS


Pi Zero 2 W / Pi 5


Camera


Depth Sensor


LoRa

ROVER


ESP32


Raspberry Pi


6WD
Moon Rover Chassis


Geared Motors
(6x)


GPS


IMU


ToF /
Ultrasonic


Camera


PIR

SOFTWARE


Python


OpenCV


YOLO

ROS 2
ROS 2

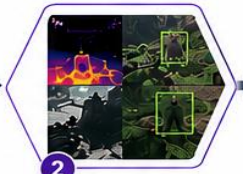

Open3D


RTAB-Map

METHODOLOGY & IMPLEMENTATION



1 **SCAN**
Drone scans the landslide area



2 **DETECT**
AI (YOLO) detects possible survivors (RGB + Thermal)



3 **LOCATE**
Location is estimated and mapped (GPS + Depth)



4 **HANDOFF**
Target coordinates sent to rover (LoRa/Telemetry)



5 **NAVIGATE**
Rover navigates autonomously (SLAM + Sensors)



6 **CONFIRM**
Close-range confirmation using camera/PIR (Human-in-loop)



7 **RESCUE**
Rescue team intervenes and provides aid

PROTOTYPE FLOW

DRONE


Aerial Scanning
(RGB + Thermal + Depth)

AI PROCESSING


YOLO Detection
& Mapping

TARGET COORDINATES


Location Sent via
LoRa / Telemetry

ROVER (6WD)


Receives Target &
Starts Navigation

CONFIRMATION


Close-range Check
(Camera / PIR)

RESCUE TEAM


Final Rescue
& Human Aid

NEXORA

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

❖ Feasibility, Challenges & Mitigation

- **Feasibility:** Low-cost • Modular • Scalable
- **Challenges:** GPS Loss • Rough Terrain • AI Errors • Battery
- **Mitigation:** Sensor Fusion • Robust Rover • AI Verification • Power Management



NEXORA

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

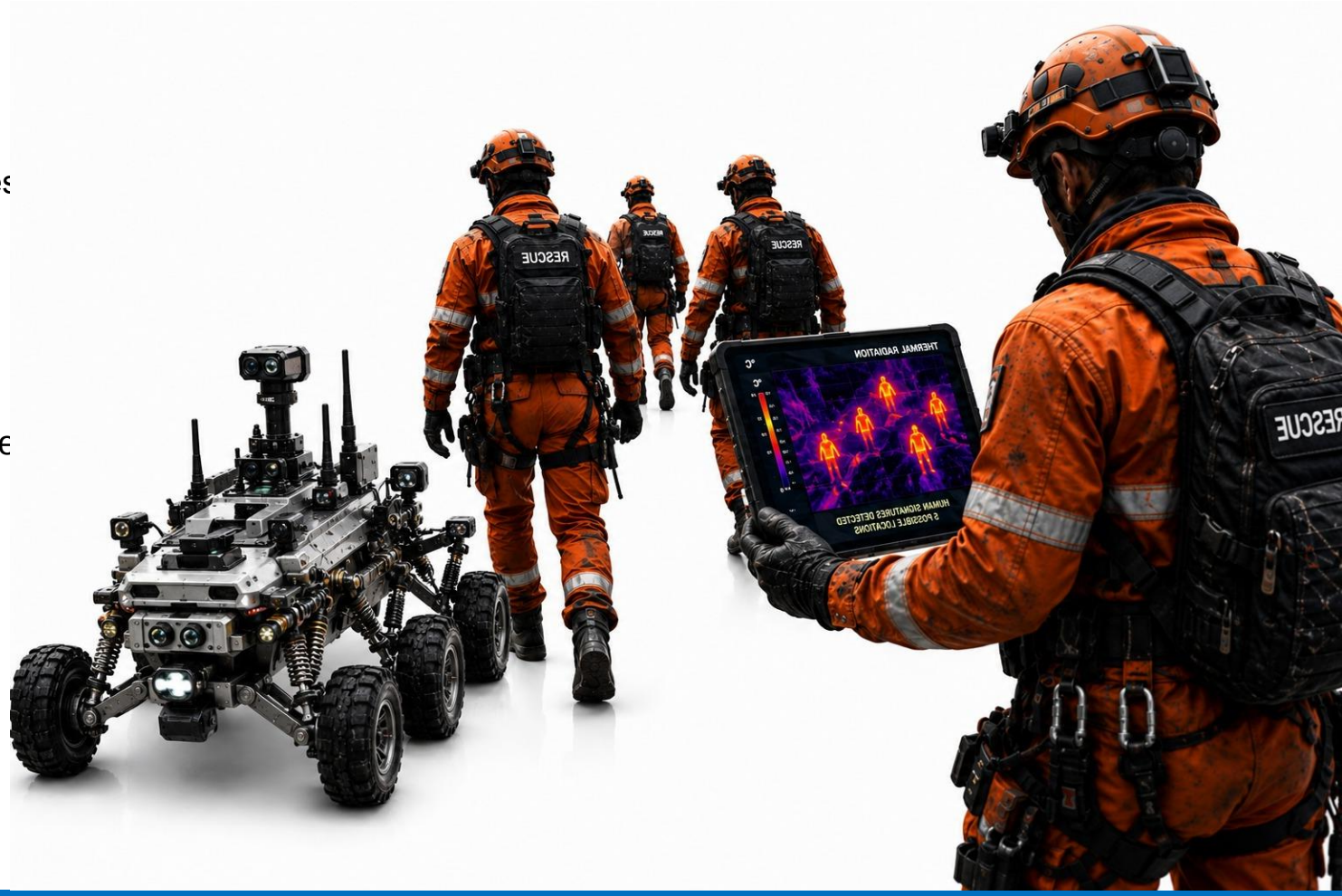
❖ Impact & Benefits:

Impact: Faster Search • Safer Rescue • Better Awarenes

Social: Protects Rescuers • Locates Survivors Faster

Economic: Low-Cost • Reusable

Environmental: Minimizes Human Exposure • Low-Powe



NEXORA

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

- **1.Huang et al.(2024)**
  FPGA-based UAV and UGV for Search & Rescue
  → UAV–UGV collaboration + RGB/Thermal + Edge AI.
https://www.researchgate.net/publication/384511724_FPGA-based_UAV_and_UGV_for_search_and_rescue_applications_A_case_study

- **2. Jadeja et al. (2024)**
  Survivor Detection for Post-Earthquake SAR
  → YOLO-based survivor detection under debris/occlusion.
https://www.researchgate.net/publication/385176704_Survivor_detection_approach_for_post_earthquake_search_and_rescue_missions_based_on_deep_learning_inspired_algorithms

- **3. Zafar et al. (2024)**
  UGV Performance in SAR Operations
  → UGV + Deep Learning + victim detection.
https://www.researchgate.net/publication/368394359_Robotics_in_Search_and_Rescue_SAR_Operations_An_Ethical_and_Design_Perspective_Framework_for_Response_Phase

- **<u>One about Our Project :</u>**

Research Gap → Practical Air-to-Ground Target Handoff